

Primers and Shortcuts: Testing Structural Preambles and Reasoning Budgets in LLM Graph Reasoning

Gal Barak

github.com/GalBrk

Nitzan Zacharia

github.com/NitzanZacharia

Inbal Moryles

github.com/iinbal

Abstract

Prepending a short *primer* of factual sentences about each node’s local structure — degree, clustering coefficient, a random-walk structural encoding, or connected-component membership — before a graph’s standard text encoding is an intuitively appealing way to ease graph-reasoning prompts for a large language model. Any apparent gain is illusory, however, if a solver can recover the answer directly from the primer text without ever consulting the graph, and it is confounded if the primer merely lengthens the prompt. We introduce a graph-blind shortcut solver as a content-only lower bound and a length-matched filler control as a length-only upper bound. Evaluated at $n = 40$ on `qwen3-1.7b` and `qwen3-4b`, with and without chain-of-thought, we find that no primer effect replicates across model scale. Across four arms, six tasks and six primers, several cells beat both `none` and `filler` significantly in one model size, but none does so in both. Nor is the filler inert: it costs the thinking arm up to 14.1 points on dense graphs and harms both sizes on `edge_existence`, and the largest apparent gains fall on tasks whose answer is effectively constant, where content-free text moves accuracy as much as a structural primer. Chain-of-thought on an identical checkpoint yields a +29.2-point pooled improvement, an order of magnitude larger than any primer effect measured. The effects that do appear are organised by one variable: a primer’s benefit falls as the model’s own accuracy without it rises, so that a primer stating the answer verbatim is worth +27.8 points where the baseline is 0.02 and −6.5 where it is 0.99 — within a single model. Our graph-blind solver separates this from a ceiling artifact: over 177 cells that offer the model a substitute route to the answer the effect tracks baseline at $r = -0.390$, while over 177 baseline-matched cells that do not it tracks nothing ($r = +0.041$). The relation replicates on eleven held-out arms spanning 0.6–14B and two model families, where

parameter count adds nothing once baseline is included ($p = 0.64$). One cell survives every control: `clustering on node_degree` for `qwen3-1.7b`, +3.8 points, replicated at +4.3 ($p = 0.0006$).

1 Introduction

Fatemi et al. (2023) demonstrate that a large language model’s accuracy on graph-reasoning questions depends substantially on how the graph is encoded as text, independent of its underlying reasoning ability. Their study established foundational baselines by varying the encoding function (e.g., adjacency lists, incident lists) while holding the question and graph fixed.

This work poses a complementary question: rather than altering how edges are described, does prepending a short *primer* of derived, node-level facts before the graph encoding help a model answer questions about that graph? While we adopt the GraphQA benchmark to evaluate these tasks, our intervention—the primer—introduces an architectural variable absent from the original suite of encoding experiments. A primer lies outside the core encoding vocabulary: it is additional, redundant information computed from the graph and expressed in plain sentences.

Two confounds make such an effect hard to read. If a primer helps because the model reads the answer out of the preamble rather than reasoning over the graph, that is answer leakage, not reasoning. Furthermore, if it helps or hurts because it lengthens the prompt, that says nothing about its *content*: Liu et al. (2024) and the needle-in-a-haystack protocol of Kamradt (2023) demonstrate that context position and length alone move retrieval accuracy.

This paper makes two contributions to isolate these confounds. First, a single, deterministic primer renderer producing seven conditions (Section 3.1) that differ only in content, allowing differ-

Figure 1 Placeholder: Route Substitution

Primer benefit against baseline accuracy, pooled over the cells where the primer states the answer: positive below ≈ 0.8 , negative above it.

Figure 1: Placeholder for the route-substitution result of Section 5.3. A primer offers a second route to the answer; whether that helps depends on whether the offered route is more reliable than the one the model already had. We report this as a behavioural regularity over 58 cells. We did not measure attention, activations or any other internal quantity, and make no claim about the mechanism inside the model.

ences between them to be interpreted as genuine content effects — among them a length-matched, information-free *filler* condition bounding the length term. Second, a *shortcut solver* (Section 3.2) that lower-bounds what a primer leaks by answering each GraphQA task using only the preamble text. We evaluate both in a density sweep on qwen3-1.7b and qwen3-4b (Section 4).

2 Related Work

We adopt the GraphQA benchmark and task definitions introduced by Fatemi et al. (2023). While their foundational work established that encoding choices (and underlying graph generators) materially affect LLM accuracy, our study shifts the focus to the impact of explicit structural preambles. To avoid confounding structural topology with task difficulty, we explicitly fix graph size and edge density (Section 4) rather than sampling structure incidentally.

A broader literature probes LLM graph reasoning capabilities. Wang et al. (2023) report that LLMs are brittle to spurious correlations in graph and problem structure; Guo et al. (2023) find that models struggle to exploit standard graph description languages directly; and Jin et al. (2024) survey the wider space of LLM-graph integration. Dziri et al. (2023) argue that transformers solve compositional tasks by linearized subgraph matching rather than systematic multi-step reasoning, raising the question a primer’s apparent benefit also raises: is a gain evidence of reasoning, or of a shorter pattern to match against? Geirhos et al. (2020) frame this generally as *shortcut learning* — a rule that performs well on a benchmark without solving the intended problem. Our shortcut solver (Section 3.2) operationalizes that concern specifically

for the primer channel.

The RWSE condition adapts the random-walk structural encoding of Dwivedi et al. (2022), developed for graph transformers; we render its return probabilities as sentences rather than feeding them to an attention mechanism. The chain-of-thought arm follows Wei et al. (2022), and our paired testing follows Dror et al. (2018).

3 Method

3.1 Primers

A primer is a short preamble of factual sentences describing a graph’s local or global structure, prepended to the standard text encoding and the task question. A single renderer produces every condition, ensuring all node-level facts share identical sentence framing (“Node N has...”) and formatting. We evaluate seven conditions. One is a baseline (**none**: the encoding with no primer). Five inject structural facts: **components**, one graph-level sentence giving the number of connected components; **degree**, each node’s exact degree; **clustering**, each node’s clustering coefficient; **rwse**, each node’s random-walk return probabilities (Dwivedi et al., 2022); and **all**, the latter three combined. The seventh, **filler**, is a content-free control of comparable length (“Node N is simply present within the graph G .”). It isolates content effects from the confound of merely lengthening the prompt.

3.2 Shortcut solvers

For a primer’s measured accuracy to constitute evidence of graph reasoning, it must exceed what is recoverable from the primer text alone. We construct a shortcut solver that combines sixteen exact rules, one heuristic rule, eight fitted rules (whose fitting and scoring graph sets are kept struc-

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946

Table 1: Shortcut ceilings: accuracy of a solver that reads only the rendered primer and never the graph, over 500 graphs from the GraphQA generator. **Bold** marks cells the primer fully determines, on which no model result can be evidence of graph reasoning.

turally disjoint), and an exact enumeration bound for small graphs. Being structurally incapable of observing the actual graph object, its accuracy on a (primer, task) cell provides a model-independent bound. Results must be read against this *shortcut ceiling* rather than against zero. Where a primer fully determines the answer — a ceiling of 1.0 — model accuracy on that cell is uninformative about graph reasoning, however high (Table 1).

3.3 Prompt assembly & Significance testing

Each prompt concatenates the primer, the graph’s text encoding, and the task question. Prompts are rebuilt from a canonicalized, parsed representation of each graph rather than reused verbatim from the source dataset, ensuring reproducible node and edge ordering.

Every comparison reported here is paired: a primer condition is compared against `none` on the same graph, with the same question and gold answer. Pairs in which either arm reached the output-token cap are dropped from that comparison rather than scored as incorrect. The primary test is the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected across the six primer conditions within a task (Benjamini and Hochberg, 1995) — $q = 0.05$. For interpreted comparisons, we additionally run paired permutation tests and a paired bootstrap 95% confidence interval on the effect size.

4 Experimental Setup

Tasks. We evaluate on six GraphQA tasks from Fatemi et al. (2023): node count, edge count, node degree, connected nodes, edge existence, and cycle check.

Models. We compare `qwen3-1.7b` and `qwen3-4b` from the Qwen3 model family (Qwen Team, 2025), each evaluated in a non-thinking

(greedy decoding, no chain-of-thought) and a thinking (chain-of-thought-enabled; Wei et al., 2022) variant, for four model arms in total. We selected the Qwen3 family for its strong architectural priors in processing structured data and code. Because Qwen3 utilizes Grouped-Query Attention (GQA) and is optimized for long-context syntactic tracking, it possesses robust native mechanisms for spatial enumeration, such as traversing an incident list. Evaluating both a base and a reasoning-enabled checkpoint allows us to observe how explicitly injected structural priors (the primers) interact with the model’s internal computational budget and its inherent topological representation.

Sweep design. All four arms are evaluated on one shared set of prompts. Graphs are Erdős–Rényi (Erdős and Rényi, 1959) with a fixed node count of 40 and edge density swept over four levels ($p \in \{0.10, 0.20, 0.35, 0.50\}$). Every combination of the seven primer conditions and six tasks is evaluated on 100 graphs per (density, task) cell, yielding 16,800 prompts per model arm. Generation is greedy, with a raised output budget (up to 8192 tokens). Auxiliary sweeps include a fixed `node_degree` sweep at 400 graphs per density level, and an extended range sweep up to $p = 0.85$.

Held-out arms. Eleven further model arms are held out as a replication set. No checkpoint among them appears in the main sweep, so neither the model nor the corpus was used to develop any claim here. Eight are evaluated on the published GraphQA split (30 graphs per cell, five tasks, six conditions): `qwen3-8b` and `qwen3-14b`, and `gemma-4-E4B-it` and `gemma-4-12B-it` from a second model family, each with and without chain-of-thought. Three more — `qwen3-0.6b` with and without chain-of-thought, and `qwen3-2b` — answer a 100-graph `{none, degree}` probe over all six tasks. Together they extend the comparison from 1.7–4B in one family to 0.6–14B in two.

Metrics and reproducibility. We use exact match for integer- and boolean-valued tasks and set-F1 for connected nodes, reporting every result against both `none` and the corresponding shortcut ceiling. Code, prompt sets, and raw generations are released with the paper.

Known validity caveats. Several cells are shortcut-dominated at this graph size rather than

genuine tests of graph reasoning: any primer with one sentence per node lets node count be recovered by counting sentences; cycle check’s gold answer is “yes” for nearly every graph past the sparsest level; and degree states the node-degree answer verbatim. The one cell established as free of this contamination is node degree under {none, components, clustering, filler}.

5 Results

All four arms completed the sweep: 16,800 scored generations each. Unless stated otherwise, every reported Δ is a paired difference against none over instances where neither arm truncated.

5.1 Truncation must be read before accuracy

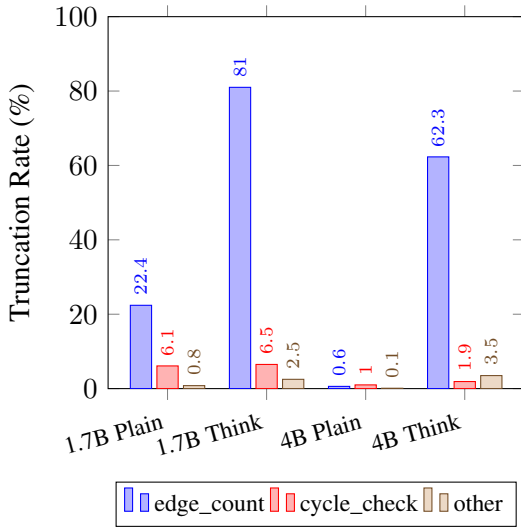


Figure 2: Truncation rate (% of generations reaching the 8192-token output cap). `edge_count` exhausts the output budget for the thinking arms.

The truncation rate (Figure 2) is neither uniform across tasks nor across arms, and it is not ignorable: on `edge_count` the thinking arms truncate on 62–81% of instances. Because truncated generations are dropped rather than scored incorrect, the survivors there are a biased subset of the sparsest graphs. We report `edge_count` under chain-of-thought in the Appendix but decline to interpret it. The choice to drop rather than zero-score truncated rows is consequential: scoring them as incorrect reverses the sign of the `rwse` effect on 1.7B `cycle_check`.

5.2 The full density sweep

No primer effect replicates across model scale. Several cells significantly exceed both none and

filler in one plain arm; none does so in both. The strongest candidate is degree, and all which contains it, on `node_degree` — the one task where a primer states the requested quantity verbatim. It yields a +7.5-point gain for qwen3-1.7b, yet penalizes qwen3-4b by −6.5 points, with non-overlapping confidence intervals between the sizes. A primer that *contains the answer* demonstrably degrades the performance of the larger model.

5.3 The effect is organised by baseline accuracy, not by scale

Reading the reversal above as an effect of model size is a coincidence of which tasks we chose. A variable that has nothing to do with parameter count explains it better: how accurate the model already was on that cell without any primer. The rest of this section tests that claim three ways — against an instrument we already have, against an experimental manipulation, and on models we did not use to formulate it.

The shortcut solver separates two kinds of primer. The graph-blind solver of Section 3.2 measures, per (task, primer), how much of the answer is recoverable from the primer text alone. That measurement is an instrument, not just a control: it splits the conditions into those that offer the model a substitute route to the answer and those that add text without one. Table 3 applies the split to the 354 cells of the four non-degenerate tasks. Where a substitute route exists, the effect declines with baseline at $r = -0.390$ ($p = 8 \cdot 10^{-8}$) and crosses zero at 0.79. Where none exists, there is no relation at all: $r = +0.041$ ($p = 0.59$), and for the content-free filler alone, $r = -0.006$ ($p = 0.97$).

This is the control that regression to the mean has to survive, and does not. Both groups hold 177 cells, drawn from the same four tasks, with matched baseline distributions; a cell at 0.99 has the same nowhere-to-go pressure in either. Only the primers that hand the model an answer show the slope.

The same pattern appears when difficulty is manipulated, not observed. Table 4 holds the arm, the task and the primer fixed and varies only edge density, which drives `node_degree` accuracy from 0.95 down to 0.44. The degree primer’s benefit rises monotonically across the seven levels, from +3.1 to +34.2 points, $r = -0.975$ ($p = 0.0002$). No sampling artifact produces a

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b	connected_nodes	72.5	+0.8	+1.2	+1.0	+3.5	-6.2	+2.0
	cycle_check	99.7	-0.6	+0.3	-7.5	-12.6	+0.3	-1.1
	edge_count	1.8	-1.4	-0.4	-0.8	+4.0	-0.4	+2.4
	edge_existence	70.5	-8.2	+3.8	+5.3	+4.0	-15.8	+11.5
	node_count	31.5	+21.2	+67.2	+2.0	+6.8	+39.8	+47.0
	node_degree	60.5	+1.8	+3.0	+4.0	+7.5	-0.5	+10.2
qwen3-4b	connected_nodes	86.0	+9.2	+1.0	0.0	-3.0	-12.5	-2.8
	cycle_check	99.7	-1.3	+0.3	+0.3	-1.5	-0.3	-0.5
	edge_count	2.0	-1.3	-1.3	-1.5	+27.7	+1.2	+3.8
	edge_existence	90.5	-2.8	+4.0	-10.0	-4.3	-7.0	-9.8
	node_count	90.2	+9.0	+8.8	+9.2	+8.0	+9.2	+9.8
	node_degree	99.2	+0.2	-0.5	-2.2	-6.5	-0.5	-8.0

Table 2: Main sweep, plain arms (thinking arms in Table 6), $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The *none* column is absolute accuracy (%); all other columns are paired differences against *none* in percentage points. Truncated generations are excluded pairwise, giving $n \in [391, 400]$ paired instances per cell. **Bold** indicates significance under an exact McNemar test after Benjamini–Hochberg correction ($q = 0.05$).

Accuracy under	substitute route		no route	
<i>none</i>	cells	Δ	cells	Δ
0.00–0.25	24	+6.3	24	-0.7
0.25–0.50	6	+8.5	6	+2.0
0.50–0.75	18	+11.1	12	-8.1
0.75–0.95	27	-1.7	17	-2.1
0.95–1.00	102	-3.3	118	-0.4
$r(\Delta, \text{baseline})$	-0.390		+0.041	
	$p = 8 \cdot 10^{-8}$		$p = 0.59$	

Table 3: Mean primer effect in percentage points, binned by the model’s accuracy on the same cell *without* a primer. A cell is one (arm, task, density, primer); the two degenerate-answer tasks are excluded. Cells are split by the graph-blind shortcut solver: on the left, primers from which that solver recovers more of the answer than it does from *none*; on the right, primers from which it recovers nothing extra. The two groups contain 177 cells each and have almost identical baseline distributions (0.785 ± 0.341 against 0.800 ± 0.342), so any ceiling pressure applies equally to both.

monotone ramp of that size against an experimentally set variable.

The same table shows the second effect, running the other way. The content-free *filler* is harmless on sparse graphs and costs -14.1 points on dense ones, $r = +0.953$ ($p = 0.0009$); the plain arm agrees ($r = +0.805$, $p = 0.029$). Added text is not a fixed tax. Its cost grows as the task gets harder, exactly where an answer-stating primer’s benefit grows. The two mechanisms are separable and have opposite signs, which is why pooling all conditions into one correlation understates both.

The relation holds on eleven unseen arms and a second model family. On the held-out arms

p	<i>none</i>	degree	clust.	filler
0.10	0.952	+3.1	+1.8	+2.3
0.20	0.877	+6.4	+4.8	+2.5
0.35	0.632	+19.6	+3.3	-4.3
0.50	0.585	+22.8	-0.8	-9.5
0.65	0.439	+29.3	-2.0	-9.3
0.75	0.442	+30.6	-8.8	-14.1
0.85	0.494	+34.2	-6.1	-11.7
r		-0.975	+0.763	+0.953
p		0.0002	0.046	0.0009

Table 4: Difficulty manipulated experimentally rather than observed. One arm (qwen3-1.7b-think), one task (node_degree), up to 400 paired graphs per level; only edge density varies. *none* is accuracy, the rest are paired effects in percentage points. Every degree entry is significant; *filler* is significant at the four densest levels. The answer-stating primer and the content-free one move in opposite directions as the task hardens. The last two rows give the correlation between each column’s effect and the *none* accuracy beside it, across the seven levels.

of Section 4 — 0.6B to 14B, Qwen3 and Gemma — the relation reappears at full strength where a substitute route exists: $r = -0.662$ over 73 cells ($p = 2 \cdot 10^{-10}$). Over the 54 cells without one it is not significant, $r = -0.206$ ($p = 0.14$), so the contrast survives, though less sharply than in Table 3; these two groups are not baseline-matched as those were (0.903 ± 0.206 against 0.949 ± 0.151), which is why Table 3 and not this comparison carries the control. Regressing the effect on baseline, log parameter count and reasoning mode together gives a baseline coefficient of -30.7 ($p = 1 \cdot 10^{-10}$) and a parameter coefficient of +0.39 ($p = 0.64$). Once baseline is in the model, scale contributes nothing

measurable. Gemma alone cannot confirm the relation — its baselines have almost no spread left (0.982 ± 0.028 , $r = -0.280$, $p = 0.14$) — but it does not contradict it either.

One quantity does not transfer: the crossover. It sits at 0.79 on the $n=40$ corpus and near 0.99 on the published split, so the threshold is a property of a corpus rather than a constant, and predicting the sign of a small effect on a new corpus from our fit does no better than chance.

Route substitution. A primer that states the answer offers a second way to produce it, and the model takes that route. Whether this helps depends on whether the offered route is more reliable than the one the model already had, which is what the baseline measures. Below the crossover, copying beats computing; above it, copying is the noisier option and the primer is a net cost. Within qwen3-4b alone the degree primer is worth +27.8 points on `edge_count`, where that arm’s baseline is 0.02, and −6.5 on `node_degree`, where it is 0.99, with no between-model comparison involved. This subsumes the non-replication result: an effect measured at one baseline need not transfer to a model at a different one.

The practical consequence is that a primer effect reported against a no-primer baseline, without reporting that baseline, is uninterpretable. The same intervention reads as +27.8 or −6.5 depending on a number the table does not otherwise print.

Answer retrieval is partial even where it is complete in principle. The shortcut solver attains 1.00 on `node_degree` under `degree` and `all`, and on `edge_count` under the same two. No arm approaches that ceiling: the largest `edge_count` gain anywhere is qwen3-4b under `degree`, +27.7 points, still leaving roughly 70% of instances wrong against a bar of 100%.

Trivially-answerable tasks respond to text, not structure. The largest apparent effects in Table 2 fall on `node_count`. As noted in Section 4, its gold answer is the constant 40 at this graph size. In both arms, the content-free `filler` is among the strongest conditions. The dominant `none` failure is the answer “39”, an off-by-one from the encoder’s “nodes 0, . . . , 39” preamble that any interposed text appears to disrupt. We retract the reading of this cell as a structural primer effect.

Prompt length is not a neutral covariate. The `filler` condition is not inert.

It significantly harms qwen3-1.7b on `edge_existence` (−15.8 [−20.0, −11.8]) and `connected_nodes` (−6.2 [−10.2, −2.2]), and qwen3-4b on `connected_nodes` (−12.5 [−17.0, −8.0]) and `edge_existence` (−7.0 [−10.8, −3.5]). Over a wider density range (p up to 0.85, 2,774 pairs, qwen3-1.7b-think), the filler costs −6.3 points pooled ($p < 10^{-4}$): nothing significant at $p \leq 0.35$, followed by sharp declines of −9.5, −9.3, −14.1, and −11.7 at the four densest levels. Length incurs a heavier penalty on dense graphs than most primer contents gain anywhere here, confirming that comparable effects measured against `none` cannot be credited to content without a length control.

Chain-of-thought dominates every primer effect.

Activating chain-of-thought reasoning, while fixing the checkpoint and prompts, yields an average improvement of +29.2 points (and +50.0 at $p = 0.85$) — an order of magnitude above the largest non-retrieval primer effect measured. The consequence is that the thinking arms sit at ceiling on `cycle_check`, `edge_existence` and `node_count` under every condition, leaving no headroom for an observable primer effect.

The one cell that survives every control. On a dedicated sweep of 400 graphs per density level, `clustering` improves qwen3-1.7b on `node_degree` by +3.8 points [+1.5, +6.2], and an independent sample reproduces it at +4.3 [+1.9, +6.6] ($p = 0.0017$ and 0.0006 ; Table 5). It is positive at every density level and attributable neither to answer leakage nor to length. However, the effect does not survive into the thinking arm, and qwen3-4b cannot test it at all, being saturated under `none`.

6 Discussion

The central negative result of this study is a failure to replicate across scale: no primer effect that clears both the `none` baseline and the `filler` control in one model size does so in the other. Three regularities account for most of the variations observed, and none align with the mechanism a primer was hypothesized to supply.

First, a primer helps reliably only where it states the requested quantity verbatim, and its sign is set by the model’s baseline accuracy rather than by its size (Section 5.3). The natural reading is route substitution: the primer supplies a second way to

produce the answer, the model takes it, and this is a gain exactly when the offered route is more reliable than the one the model already had. At 1.7B, whose own enumeration of a dense incident list is unreliable, the text shortcut is an improvement; at 4B, whose enumeration is already correct on 99% of instances, it is a substitution of a noisier route for a better one.

We emphasise that this is a claim about behaviour, not about circuits. We measured no attention weights, activations or internal states, and the data place at least one limit on any account framed as the larger model *rejecting* the primer: `qwen3-4b-think` answers correctly on all 29 of the instances the degree primer breaks for `qwen3-4b`, under that identical primer and identical weights. What the primer disrupts is therefore recoverable within the same network once a reasoning trace is permitted, which is not what an interference account would predict.

Second, added text acts as an active treatment regardless of its content. The length-matched `filler` significantly harms both model sizes on `edge_existence` and `connected_nodes` and significantly helps both on `node_count`, at magnitudes exceeding nearly every content effect measured here. An experimental design that compares a primer only against `none` estimates content and length confounded, with the length term frequently emerging as the larger of the two.

Third, the headroom in which any of this could be measured is largely consumed by the decoding mode. Chain-of-thought moves most (task, arm) cells to ceiling, and where it does not saturate a task it censors it instead. It also changes how long the model generates before answering — a measurable effect on the process rather than on accuracy. For `qwen3-1.7b-think`, degree raises the non-termination rate by 3.4 points over `none` ($p < 0.001$) while the content-free `filler` lowers it, and outside `edge_count` that primer produces 137 non-terminating generations of which only 7 also fail to terminate under `none`. These are new failures attributable to the primer, not to the instance. We report the effect at that level and do not infer a termination mechanism from it: response length is an observable, the policy that produced it is not.

7 Future Work: Mechanistic Interpretability

The behavioural regularities reported here—a primer effect whose sign tracks baseline accuracy, and a condition-dependent shift in how long the thinking arms generate—are consistent with several internal accounts and identify none of them. That is a mandate for mechanistic interpretability. We plan to apply activation patching and causal abstraction to Qwen3’s grouped-query attention heads during these tasks, intervening on the heads that read the incident list while a primer is present, to test whether the substitution we observe behaviourally corresponds to an identifiable change in how the answer is computed.

Two cheaper experiments come first, because they discriminate between behavioural accounts without any interpretability machinery, and both are prompt-only changes against the existing corpus. A *single-fact* primer stating only the queried node’s degree keeps the answer available while removing the other 39 sentences, separating interference from route substitution. A *corrupted* primer stating a wrong degree measures reliance directly: a model that copies collapses, a model that computes is unaffected, and the resulting reliance curve is the quantitative form of the prediction in Section 5.3. Extending the same prompt file down and up the Qwen3 ladder (0.6B, 8B, 14B) would turn the two-point sign reversal into a curve, and because 0.6B to 1.7B varies width at constant depth while 1.7B to 4B adds depth, it would also separate those two axes. Whether the pattern holds outside one model family remains open.

8 Conclusion

We evaluated seven primer conditions across six GraphQA tasks, four densities, and four model arms at $n=40$, totaling 67,200 generations. Two controls rigorously bound the result: a graph-blind solver measuring answer retrieval, and a length-matched, content-free primer measuring prompt length. No primer effect clears both controls across both model sizes. One exception survives every control and an independent replication: clustering on `node_degree` for `qwen3-1.7b` (+3.8 and +4.3 points). We interpret the broader result not as evidence that structural preambles cannot help, but that most (model, task) pairs at this scale afford no interpretable test of whether they do. The effects that do appear

are organised by baseline accuracy rather than by scale (Section 5.3), a dissociation that our graph-blind solver isolates and that thirteen held-out arms across two model families reproduce: a primer stating the answer is worth +27.8 points where the model’s own accuracy is 0.02 and −6.5 where it is 0.99, within a single model, with the sign crossing near 0.8. The practical reading is that prepending a local fact substitutes one route to the answer for another, and is a gain only while the model’s own route is the worse of the two.

Limitations

The main sweep covers only two model sizes (1.7B and 4B parameters) from a single model family; the held-out arms extend the range to 0.6–14B and a second family, but on smaller corpora and fewer conditions, so the findings may not generalize to substantially larger models or to other architectures. The baseline relation of Section 5.3 is measured against each arm’s own accuracy, and substituting an independent difficulty estimate — the mean `none` accuracy of the other three arms on the same cell — leaves no significant relation ($r = +0.113$, $p = 0.13$). We therefore read the effect as governed by a model’s own competence on an item set, not by any difficulty intrinsic to the items. Several cells are known in advance to be shortcut-dominated at this graph size and density range; results on those cells bound what can be concluded about genuine graph reasoning even though we report them. Graphs are drawn from a single generator family (Erdős–Rényi). Whether primer effects transfer to other generators remains untested. Finally, the length control is a bracket rather than an exact match: the `filler` condition is longer than `degree` and `clustering` and shorter than `rwse` and `all`, so it bounds the length term without eliminating it. A design holding rendered length exactly constant while varying structure would isolate content more sharply, and we regard that as the natural next step.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data, and no annotation labour. We see no direct risk of harm arising from the findings. The principal societal cost is the compute expended.

References

- Yoav Benjamini and Yosef Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1383–1392.
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2022. Graph neural networks with learnable structural and positional representations. In *International Conference on Learning Representations (ICLR)*.
- Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Paul Erdős and Alfréd Rényi. 1959. On random graphs i. *Publicationes Mathematicae Debrecen*, 6:290–297.
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2023. Talk like a graph: Encoding graphs for large language models. *arXiv preprint arXiv:2310.04560*.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673.
- Jiayan Guo, Lun Du, and Hengyu Liu. 2023. Gpt4graph: Can large language models understand graph structured data? an empirical evaluation and benchmarking. *arXiv preprint arXiv:2305.15066*.
- Bowen Jin, Gang Liu, Chi Han, Meng Jiang, Heng Ji, and Jiawei Han. 2024. Large language models on graphs: A comprehensive survey. *IEEE Transactions on Knowledge and Data Engineering*, 36(12):8622–8642.
- Greg Kamradt. 2023. Needle in a haystack – pressure testing llms. https://github.com/gkamradt/LLMTest_NeedleInAHaystack.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35.

A Supplementary tables

Sample	Δ	n	95% CI	p
Original	+3.8	1600	[+1.5, +6.2]	0.0017
Replication	+4.3	1600	[+1.9, +6.6]	0.0006

Table 5: qwen3-1.7b, node_degree, clustering against none, $n=40$, pooled over $p \in \{0.10, 0.20, 0.35, 0.50\}$ with 400 paired instances per level. Δ in percentage points; n is paired instances, the interval is a paired bootstrap on the effect size, and p is the exact McNemar test. The replication draws a fresh graph sample at the same design points. The shortcut ceiling for this cell is 0.08, so the effect is not answer retrieval, and clustering is 200 characters shorter than the filler control, which registers no effect on this task in either 1.7B arm.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b-think	connected_nodes	79.7	+2.8	+1.5	−0.3	+7.4	+2.0	+6.1
	cycle_check	100.0	0.0	0.0	0.0	−0.7	0.0	0.0
	edge_count [†]	86.3	+7.4	−8.7	−4.0	−61.5	+3.4	−8.3
	edge_existence	99.2	−0.8	−1.0	−1.3	−2.3	−0.8	−2.0
	node_count	98.6	+0.9	+1.4	+1.4	+0.8	+1.4	+1.4
	node_degree	76.8	−1.0	+1.0	+2.5	+11.7	−3.5	+11.6
qwen3-4b-think	connected_nodes	98.7	+0.5	0.0	0.0	0.0	+0.8	−0.5
	cycle_check	100.0	0.0	0.0	0.0	−0.8	0.0	0.0
	edge_count [†]	97.6	+1.7	+1.7	0.0	−23.5	−2.0	−41.5
	edge_existence	100.0	0.0	0.0	0.0	0.0	0.0	0.0
	node_count	99.5	0.0	+0.5	+0.5	+0.5	+0.5	+0.5
	node_degree	99.7	−0.3	−0.3	−2.1	0.0	0.0	−0.8

Table 6: Main sweep, thinking arms; conventions as in Table 2. Both arms are at or near ceiling on most tasks, leaving no headroom for a primer effect. [†]Censored beyond interpretation; $n \in [13, 60]$ surviving pairs.